

Tide retrospective: the forward theorems the germbij forward direction closes

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Abstract

The public face of the forward-expansion programme, and its close: on a `ForwardExpansionDomain` — a nondegenerate local minimum with C^{N+2} regularity and the Peano remainder — the rescaled posterior moment of every continuous observable of polynomial growth is an order- N asymptotic polynomial at 0^+ (`rescaledMoment_hasExpansion`). The coefficients are the division of the numerator expansion by the partition expansion, whose constant coefficient $\int e^{-T_2}$ is positive; the monomial instances are exactly the moment families that the merged inverse-half theorems consume; and moment families agreeing to $o(q^N)$ have equal coefficients through order N . Together with the inverse half — superpolynomially-close localized moment families determine the location, the full positive-order jet, and for analytic losses the germ — this completes the germbij note’s Theorem 3.1 in the nondegenerate case: expansion data and Taylor data determine each other, in both directions, in Lean.

1 Setting

Stage 7 of the seven-stage design consult, which ran start to finish as a single stacked arc: the expansion predicate and coefficient uniqueness; the mesoscopic window and its tail calculus; the one-field Peano mixin and the exact exponent split; the graded exponential; the coefficient functions; the scalar bounds; the quadratic bridge and Gaussian absorption; the q -uniform majorant; the window dominated convergence; the outer tails and the numerator packaging; asymptotic division; and now the public statements. Each stage’s interface was consumed by the next with essentially no rework — the two architecture consults (the programme design and the stage-5 shape) did what they were commissioned to do.

2 The thing formalised

`numeratorCoeff_one_zero_pos`: the partition constant coefficient is $\int e^{-T_2} > 0$, from the zeroth coefficient function being constantly one, the support of a positive integrand being everything, and the volume measure being nonzero. `momentCoeff` and `rescaledMoment_hasExpansion`: the division of the two numerator expansions, applied at the seabed’s `rescaledMoment`. `rescaledMoment_expansion_exists` (the existence form), `monomial_moment_hasExpansion` (the `monomialTest` instances), and `momentCoeff_eq_of_isLittle0` (the comparison face, through stage A1’s uniqueness).

3 Where progress stalled

One rename: `IsLittle0.neg` is spelled `IsLittle0.neg_left`. The tide compiled on the first fix round — fitting for a stage that was designed to be pure composition.

4 Mathlib lookup versus new mathematics

Lookup: the support characterization of positive integrals (`integral_pos_iff_support_of_nonneg`), `Set.eq_univ_of_forall`, `Function.mem_support`, `Measure.measure_univ_pos`, `NeZero.ne`. New: none. The mathematics of this tide is the programme it closes.

5 What the next tide inherits

Process work first: the blue-dot `\leanref` marker pass on the `germbij` note for the forward theorems, the SRI docs sync, and the commit-pin bump. Mathematically, what remains of the `germbij` note beyond the user's stated core: the semi-quasi-homogeneous constructive class, and — if explicit coefficient values are ever wanted rather than existence — the connection of `momentCoeff` to Wick/Isserlis data through the Gaussian moment integrals already in the seabed.